
Project Title: Script-Controlled ACL – Restrict Record Access Based on Field Value

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Team ID: SWTID-2026-2246

PHASE 5 – PROJECT DEVELOPMENT

1. Introduction:

The Project Development phase describes the actual implementation of the proposed access-control system in ServiceNow.

The project was developed using a custom table, user roles, ACL rules, a server-side script, and a branch-based data condition.

2. ServiceNow Environment Setup:

The ServiceNow instance was opened using a web browser with administrator access.

The development environment was prepared to create:

- o Users
 - o Roles
 - o Custom table
 - o Custom fields
 - o Sample records
 - o Access Control Lists
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3. User Creation:

A test user was created for the EEE branch.

User Details

Field	Value
First Name	EEE
Last Name	User
User Name	EEE User
Email	eeeuser@gmail.com

This user was used to test the access-control rules.

4. Role Creation:

Four custom roles were created:

- bb1
- bb2
- bb3
- bb4

Each role was associated with a specific operation.

Role	Permission
bb1	Read
bb2	Create
bb3	Write
bb4	Delete

The administrator role provides complete access.

5. Custom Table Creation:

A custom table named **Institution Details** was created.

Table Details

Label: Institution Details

TableName: u_institution_details

The table was created without extending another table.

6. Field Configuration:

The following fields were added to the Institution Details table:

Field Name	Type
Student Roll Number	Auto Number
Student Name	Reference – User
Faculty Name	Reference – User
Branch	Choice
Email	String
Phone Number	String
Description	Multi String

7. Branch Configuration:

The **Branch** field was configured as a Choice field.

The following values were added:

- EEE
- ECE
- CSE

These values were used to create different test records and verify the branch-based access restriction.

8. Test Record Creation:

Sample records were created in the Institution Details table.

Records were created with different branch values:

o EEE o
ECE o
CSE

These records were required to verify whether the ACL correctly restricts access to EEE records.

9. Read ACL Configuration:

A Record-level **Read ACL** was created for:

Table: u_institution_details

Operation: Read

Role: bb1

Active: Yes

Advanced: Yes

Data Condition

The following condition was configured:

Branch is EEE

This condition restricts the applicable records to the EEE branch.

10. Read ACL Script:

The following script was configured:

```
(function () {  
  
    //Allow admin users full access  
    if(gs.hasRole('admin')) {  
        return true;  
    }  
  
    //Allow users with bb1 role  
    if(gs.hasRole('bb1')) {  
        return true;  
    }  
  
    //Deny access for all others  
    return false;  
})();
```

The script checks whether the current user has the **admin** or **bb1** role.

The **Branch is EEE** data condition provides the branch-based record restriction.

11. Create ACL Configuration:

A Record-level **Create ACL** was created.

Table: u_institution_details

Operation: Create

Required Role: bb2

Users with the bb2 role can create records according to the configured access rules.

12. Write ACL Configuration:

A Record-level **Write ACL** was created.

Table: u_institution_details

Operation: Write

Required Role: bb3

Users with the bb3 role can modify permitted records.

13. Delete ACL Configuration:

A Record-level **Delete ACL** was created.

Table: u_institution_details

Operation: Delete

Required Role: bb4

Users with the bb4 role can delete permitted records.

14. Complete ACL Structure:

The final ACL configuration is:

Operation	Role	Additional Condition
Read	bb1	Branch is EEE
Create	bb2	None
Write	bb3	None
Delete	bb4	None

Administrators have full access.

15. Access-Control Implementation:

The implemented access flow is:

User → Role Verification → ACL Check → Branch Condition → Access Decision

If the required conditions are satisfied, access is granted.

If the conditions are not satisfied, access is denied.

16. Development Result:

The ServiceNow project was successfully configured with:

- Custom Institution Details table
- Required fields
- EEE, ECE, and CSE branch values
- Custom roles
- Read ACL

- Create ACL
 - Write ACL
 - Delete ACL
 - Script-controlled access
 - Branch-based Read restriction
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17. Conclusion:

The Project Development phase implemented the planned access-control system in ServiceNow. The system combines **roles, ACLs, scripts, and a Branch data condition** to control access to Institution Details records.

The completed implementation is ready for the **Project Testing** phase.
